Bellabeat Case Study: Smart Device Usage To Improve User Engagement and Retention

Executive Summary

01. Overview

This case study analyzes Fitbit smart device data as a proxy for Bellabeat users to understand daily activity patterns and identify engagement opportunities. Using SQL and Tableau, I examined trends over time, weekday differences and inactivity signals. The findings were presented in a dashboard and converted into recommendations to improve user engagement and retention.

02. Key Findings

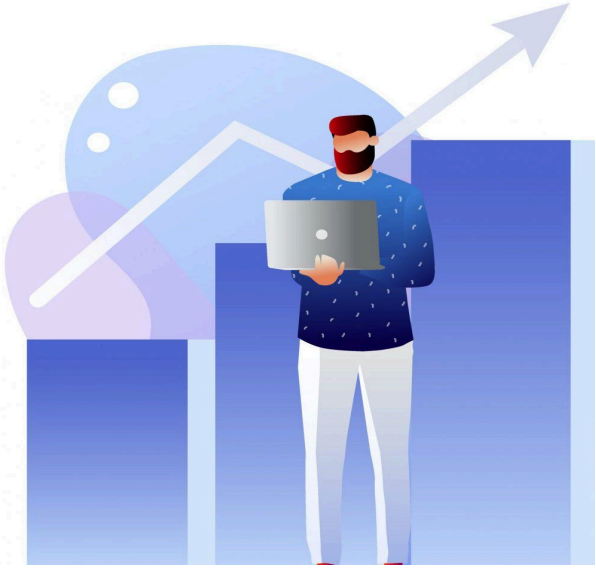
A. Engagement is inconsistent: User activity fluctuates significantly across the observed period, suggesting irregular usage patterns rather than steady daily engagement.
B. Significant inactivity exists: The dataset contains 61 user-day records with zero recorded steps, indicating potential non-usage of the device or periods of disengagement.
C. Activity varies by weekday: Average steps differ across the week, suggesting routines influence engagement and creating opportunities to time reminders or campaigns more effectively.

03. Solutions

A. Re-engagement reminders: Use app notifications to nudge users after inactivity and encourage returning to daily tracking (e.g: 1-2 days with no steps)

B. Personalized weekly nudges: Adapt messaging to routines (weekday/weekend patterns) to send prompts when users are most likely to respond.

C. Gamification & goal support: Add streaks, achievable daily goals and progress feedback to reinforce consistent usage to reduce drop-off.



04. Expected Business Impact

A. Increased daily engagement and more consistent tracking behavior.

B. Stronger perceived value of the Bellabeat app/device through personalized guidance.

C. Better-targeted marketing & product decisions by aligning campaigns with user routines.

Introduction

01. Background

Bellabeat is a wellness technology company that designs smart products focused on improving women's health and lifestyle habits. The company collects data from its smart devices to help users better understand their daily activity, sleep patterns and overall well-being.

As a growing company in the competitive health technology market, Bellabeat aims to leverage data analytics to gain deeper insights into user behavior and enhance its marketing strategies. By analyzing smart device usage data, the company seeks to better understand how consumers interact with health-tracking technologies and identify opportunities to improve customer engagement.

This case study analyzes non-Bellabeat smart device data to explore usage trends and behavioral patterns. The objective is to generate data-driven insights that can support Bellabeat's marketing decisions and help the company expand its presence in the global wellness market.

02. Objectives

A. Analyze smart device usage patterns to understand how consumers interact with health and wellness technology.

B. Identify trends in daily activity, sleep and calorie expenditure to uncover behavioral insights that can inform Bellabeat's product strategy.

C. Evaluate user engagement patterns to provide data-driven recommendations that support Bellabeat's marketing and growth decisions.

03. Problem

Bellabeat aims to expand its presence in the competitive smart wellness market but lacks a clear understanding of how consumers use smart devices in their daily lives. Without data-driven insights into user activity, sleep patterns and engagement behaviors, it is challenging for the company to optimize marketing strategies and align product decisions with customer needs.

This project was motivated by the need to transform available usage data into actionable business insights that support strategic growth.

Recommendations

1. Focus marketing efforts on active and fitness-motivated users: Bellabeat should prioritize marketing campaigns targeting users who already demonstrate consistent activity habits. These users are more likely to engage with wellness features and adopt premium services, increasing customer retention and brand loyalty.

2. Develop personalized wellness insights and reminders: Bellabeat should introduce personalized notifications based on user activity, sleep and wellness patterns. Tailored insights can encourage daily engagement and help users better understand their health behaviors.

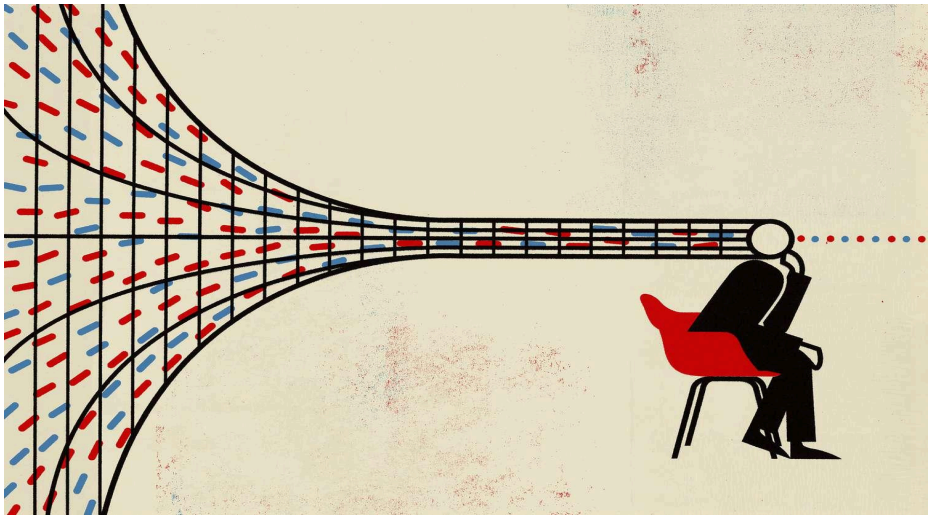
3. Improve onboarding and user education: Bellabeat should enhance onboarding experiences by providing clear guidance on how to use device features effectively. Educating users early can improve long-term engagement and reduce user drop-off.

Impact of Recommendations

The following assessment outlines the expected effectiveness and business impact of each proposed recommendation based on the analysis findings.

SOLUTION	EFFECTIVENESS	IMPACT	NOTES
Focus marketing on active users	Highly Effective	Higher engagement & retention	Targets users already showing strong activity patterns
Personalized wellness insights	Highly Effective	Increased daily usage & habit formation	Encourages consistent interaction with device
Improve onboarding & education	Moderately Effective	Reduced churn & better feature adoption	Helps new users understand product value

Analysis



01. Research Methods

This case study uses a quantitative data analysis approach based on secondary data collected from Fitbit smart device users. The dataset, sourced from Kaggle contains anonymized daily activity, sleep and health tracking information from multiple users over a defined period.

The analysis focuses on identifying behavioral patterns and usage trends through data cleaning, transformation and exploration using SQL and data visualization tools. This method enables the extraction of actionable insights to support data-driven business recommendations for Bellabeat.

02. Approaches Used

The analysis followed the structured data analysis process: Ask, Prepare, Process, Analyze, Share and Act.

Data was prepared and cleaned using SQL to ensure consistency and reliability across multiple datasets. Exploratory data analysis was conducted to identify trends in user activity levels, sleep behavior and device engagement. Visualization techniques were applied using Tableau to communicate insights clearly and support business decision-making.

This framework ensured a systematic and reproducible analytical workflow aligned with industry best practices.

03. Relevant Facts and Information

- The dataset includes daily activity and sleep data from approximately 30 Fitbit users.
- Users demonstrate higher engagement during weekdays, suggesting structured routines influence device interaction.
- Highly active users show more consistent device usage, reinforcing the opportunity to target fitness-motivated segments.
- Sleep tracking engagement varies significantly, highlighting inconsistent use of wellness monitoring features.
- The dataset required cleaning and integration across multiple files, and the sample size may limit generalizability.

Conclusion

01. Summary of Findings

This case study examined smart device usage data to better understand user behavior patterns and identify opportunities for Bellabeat to increase engagement.

The analysis showed that highly active users demonstrated more consistent device usage, highlighting a strong connection between fitness motivation and sustained product interaction.

Engagement levels also varied across weekdays and wellness features. While activity tracking was widely adopted, sleep tracking showed inconsistent usage patterns. These insights reveal opportunities for Bellabeat to enhance engagement through targeted marketing, personalized wellness insights and improved onboarding experiences.

02. Implications of the Study

The findings suggest that Bellabeat can strengthen user retention by focusing on behavior-driven engagement strategies. Prioritizing highly active users while supporting less engaged users through personalized insights and improved onboarding can increase long-term product adoption.

By leveraging behavioral data to guide marketing and product decisions, Bellabeat can enhance customer satisfaction, improve daily usage consistency and reinforce brand loyalty.

Although the dataset was limited in size, the observed patterns provide meaningful directional insights for future data-driven initiatives.

"Data-driven insights transform user behavior into strategic growth opportunities."

References

- Fitbit Fitness Tracker Data. Kaggle Dataset. https://www.kaggle.com/datasets/arashnic/fitbit
- Google Data Analytics Professional Certificate – Bellabeat Case study Materials. Coursera.